Methods to Measure Weight on Instrumented Cantilever Beam

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***Abstract*—This experiment utilized a cantilever beam instrumented with a strain gauge to measure the weight of a soda can using two different methods. The strain gauge was wired to a Wheatstone Bridge (WSB) and amplifier that converted the resistance changes into amplified voltages. A Data Acquisition device (DAQ) measured these voltages and sent them to a LabVIEW program that calculated the weight on the beam with two different methods. The first was called the beam scale method which relied on mechanics of materials principles to calculate the weight applied. The second was the calibration method which involved the calculation of a constant that related voltage change to weight on the beam. These two methods were compared by measuring the weight of the full soda can on the beam as well as the weight of the soda can as gulps were incrementally taken from it. After ten initial weight measurements of the soda can were taken, the beam scale method recorded a mean weight of 4.01 ± 0.15 N while the calibration method recorded a mean weight of 3.77 ± 0.14 N. A comparison to the weight of the full can measured on a digital scale, 3.7905 ± 0.0005 N, revealed that the calibration method was more accurate than the beam scale method. For the gulp experiment, thirteen gulps were taken to empty the can and the mean gulp size from the beam scale method was equal to 0.28 ± 0.16 N while the mean from the calibration method was equal to 0.26 ± 0.15 N. While the calibration method was deemed to be more accurate, an uncertainty analysis of the weight of the soda can during the gulp experiment revealed that it was not the more precise method.**

***Index Terms*— Calibration, Cantilever Beam, Strain Gauge, Wheatstone Bridge**

I. Introduction

THE main objective of this lab was to use a strain gauge as a transducer to measure the weight of a soda can, and the gulps taken from the can, placed at the free-end of a cantilever beam [2]. Two different methods were used to measure the weight of the soda can after each gulp. The first, called the beam scale method, relied on mechanics of materials principles to solve for the weight applied on the beam. The second method involved developing a calibration constant to calculate the weight on the beam based off a linear relationship with the amplified voltage change. Both these methods were tested and their uncertainties were compared to determine which one was the best for accurately measuring weight on a cantilever beam.

A transducer is a device that changes one physical phenomena into another. For this lab, a strain gauge mounted on a cantilever beam acted as a transducer by producing a change in resistance from a flexural load that stretched the gauge. The strain gauge was wired to a Wheatstone Bridge and amplifier to convert the resistance to a voltage. A WSB is constructed of two simple circuits wired in parallel called a voltage divider [2]. A quarter WSB utilizes two voltage dividers connected in parallel and makes a differential voltage measurement between the two dividers to calculate an unknown resistance where all other resistances are known, as shown by the configuration in Fig. 1 below [2].

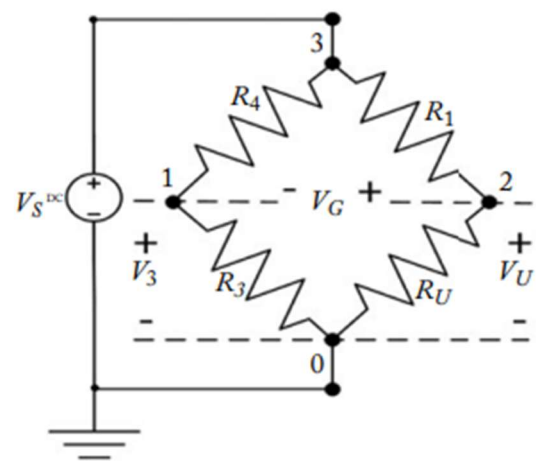

Fig. 1. Two voltage dividers connected in parallel to form a quarter Wheatstone Bridge [2].

Through the above configuration, the WSB can convert the resistance change from the strain gauge to a voltage, V_g. The amplifier then applies a gain to increase V_g to an amplified voltage, V_{amp}. For this lab, only the amplified voltage was relevant for data collection. The voltage generated by the WSB and amplifier was measured by a Data Acquisition device (DAQ) with an Analog to Digital Converter (ADC). These voltages were then sent from the DAQ to the LabVIEW to make calculations to solve for the weight on the beam. Correct sample windows had to be selected for the excitation voltage of the bridge, V_s, and the amplified voltage, V_{amp}, to ensure that these voltage values were within the range that the ADC could approximate to prevent saturation from occurring. The resolution of the ADC was optimized by choosing the smallest sample window where saturation did not occur.

The mechanics of materials principles used to calculate weight in the beam scale method stems from basic cantilever beam theory. The dimensions of the beam and the position of the strain gauge and weights are detailed in Fig. 2 below where L_o is the beam length, h is the height, b is the width, and L is the top surface distance from the center of the strain gauge to the applied weight, W [2].

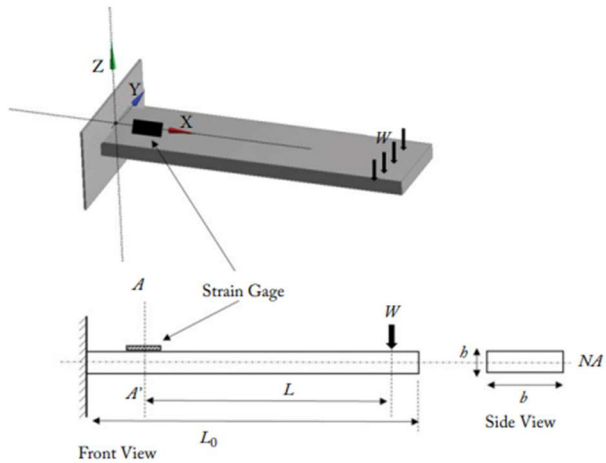

Fig. 2. Cantilever beam diagram with a strain gauge mounted on its surface and a concentrated load acting at its free end [2].

The flexural stress in a cantilever beam along the x axis and through its cross section is given by (1) below where σ_x is the flexural stress in the x direction, M is the bending moment about the y axis, y is the distance from the center line of the bar to the surface (in this case) and I is the moment of inertia [2].

$$\sigma_x = \frac{My}{I} \tag{1}$$

Equations for M, y and I based on the variables defined in Fig. 2 are shown below (2), (3) and (4).

$$M = WL \tag{2}$$

$$y = \frac{h}{2} \tag{3}$$

$$I = \frac{bh^3}{12} \tag{4}$$

Substituting (2), (3) and (4) into (1) yields an equation for flexural stress in the x direction in terms of the cantilever beam dimensions and concentrated load at the free end (5).

$$\sigma_x = \frac{6WL}{bh^2} \tag{5}$$

Although the stress is acting on the x axis and the load, W, is acting downward along the y axis, it can be assumed that this is the only stress acting on the beam [2]. Therefore, 1D Hooke's law can be used to determine strain in the axial direction, ε_x, by using the modulus of elasticity, E, as shown in (6).

$$\sigma_x = E\varepsilon_x \tag{6}$$

The beam scale method used these mechanics of materials equations to determine the weight applied at the end of the beam. The values for b, h, and L were measured with digital instruments and inputted into a subVI along with E which was provided. The strain, ε_x, was estimated from the quarter bridge strain gauge equation shown below in (7), where ΔV_{amp} is the change in amplified voltage, *gain* is the ratio of the amplified voltage to the input voltage, V_s is the excitation voltage of the bridge, and G_f is the gauge factor.

$$\varepsilon_x = \frac{4\Delta V_{amp}}{gainV_sG_f} \tag{7}$$

To calculate weight from the previously defined equations, (6) was first substituted into (5) which was then rearranged to solve for W as shown in (8) below.

$$W = \varepsilon_x(E\frac{bh^2}{6L}) \tag{8}$$

Finally, (7) was substituted into (8) to get an equation for weight in terms of all input parameters or variables measured by the DAQ as shown in (9) below.

$$W = (\frac{4\Delta V_{amp}}{gainV_sG_f})(E\frac{bh^2}{6L}) \tag{9}$$

The LabVIEW program used the above equation to calculate the weight on the beam for the beam scale method based on mechanics of materials principles. On the other hand, the calibration method only utilized the amplified voltage sent from the DAQ and a constant to approximate the same weight on the beam. The calibration constant was determined by weighing brass slotted weights on both a digital scale and the beam scale. The weights from the digital scale were then plotted with the respective amplified voltages from the beam scale data. The calibration constant was equal to the slope of this graph which was the change in weight over the change in amplified voltage. Therefore, the calibration method approximated the weight applied at the free end of the beam using (10) below where W_c is the weight approximated from the calibration method and C is the calibration constant.

$$W_c = C\Delta V_{amp} \tag{10}$$

The beam scale method based on mechanics of materials (9) and the calibration method (10) were both used to measure the weight of the soda can and the size of each gulp taken from the can. The weight calculated from each method had a different value and uncertainty due to the different contributing variables. The beam scale method had many given and statistical uncertainties contributing to the uncertainty of the weight, while the uncertainty of the weight approximated from the calibration method was largely dependent on the uncertainty

of the calibration constant which was determined from a Monte Carlo simulation. An uncertainty analysis for both methods was conducted to determine the most accurate and precise approach to determining the weight applied to the free end of the instrumented cantilever beam.

II. Procedure

The required materials for this lab were a DAQ, WSB and amplifier, laptop capable of running LabVIEW, a digital scale, 6061 T6 aluminum bar, cantilever beam mount, strain gauge, 2 conductor strain gauge wire, solder iron and wire, clear tape, masking tape, glue, sandpaper, alcohol wipes, pencil, ruler, digital calipers, digital micrometer, wire stripper, snap knife, wood block, multimeter, screwdriver, brass slotted weights, and a soda can. The first part of this lab procedure was to construct an instrumented cantilever beam by gluing the strain gauge on the aluminum bar and securing it in the mount. Additionally, the strain gauge wires had to be soldered to jumper wires to allow for connection to the WSB and DAQ as part of the initial set up phase. The experimental part of the procedure involved using brass slotted weights to determine a calibration constant, initial weight measurements of the full soda can, and measuring gulp sizes using the beam scale method and the calibration method. Both the initial setup and experimental procedures are detailed below.

Initial Setup Procedure

The first step of constructing the instrumented cantilever beam was to prepare the surface of the aluminum bar. A strip of sandpaper was used to remove paint and surface irregularities on the area of the bar about four centimeters from the mount end as this is where the strain gauge would be placed [3]. The surface of the bar was then cleaned with an alcohol wipe to remove the debris from the sandpaper and reveal the smooth surface. The final preparation step was to use a ruler to measure to the four-centimeter mark and wrap a piece of masking tape around the bar at this point. This was done to properly position and align the strain gauge on the bar as will be seen in the upcoming steps.

It was important to not touch the strain gauge as human oils would possibly impact its functionality. Therefore, a piece of clear tape was used to pick up the strain gauge. The strain gauge was aligned and centered right in front of the masking tape on the side of the beam that would be mounted. The side of the strain gauge with the leads was positioned towards what would become the free end of the beam. Once the strain gauge was aligned correctly, the half of the tape on the side of the strain gauge with the leads was pressed down. The other half of the tape was not pressed down so it could be lifted to apply glue under the strain gauge. The tape was carefully lifted on this side and a small dot of glue was applied directly under the strain gauge. The lifted side of the tape was then finger pressed down and smoothened to ensure the glue was evenly applied through the area. The beam was then set aside to allow the glue to dry.

While the glue was drying, the wires that would be connected to the strain gauge leads were prepared. A segment of 2 conductor strain gauge wire about 45 centimeters in length was obtained [3]. A handheld wire stripper was then used to remove approximately one centimeter of insulation from both ends of the wire. Next, the wire was placed on the wooden block and the blade of the snap knife was aligned at the connection point of the two wires about four centimeters from the end [3]. The knife was then pressed down and slowly cut through the connecting region of the wires to separate them along the four-centimeter distance. The ends of the wires were then pulled apart to ensure the knife had fully cut through. This was done for both ends of the wires as it would make for easier soldering to the strain gauge leads and easier connection to the strain gauge terminals on the amplifier breadboard.

Enough time had now passed for the glue to completely dry. The clear tape was gently removed by starting at the corner and pulling in a diagonal direction aligned with the corner of the strain gauge. The speed of the peeling began slow but then increased in speed when the position of the strain gage was approached. This removal method ensured the strain gauge and leads were not disturbed. Next, the dimensions of the bar were measured and recorded. The width of the bar was measured with a digital caliper and the thickness was measured with a digital micrometer near the free end. Before the relevant length measurement could be taken, the diameter of the soda can had to be measured first. A pencil was used to make a horizontal tick mark at distance equal to half the can diameter away from the free end of the bar. Finally, the length was measured with a digital caliper from the center of the strain gauge to the tick mark near the free end of the bar. All these measurements, along with the provided modulus of elasticity of the bar, were input into a SubVI of the LabVIEW program and saved as the default values.

The next step of the set up phase was to position and secure the wires to the aluminum bar. The separated ends of the wires near the strain gauge leads were bent upwards to make for easier connection. A strain relief hump was created two centimeters further down the wire [3]. Masking tape was wrapped around the bar to secure the wire at the start and end of the hump as shown in Fig. 1 below.

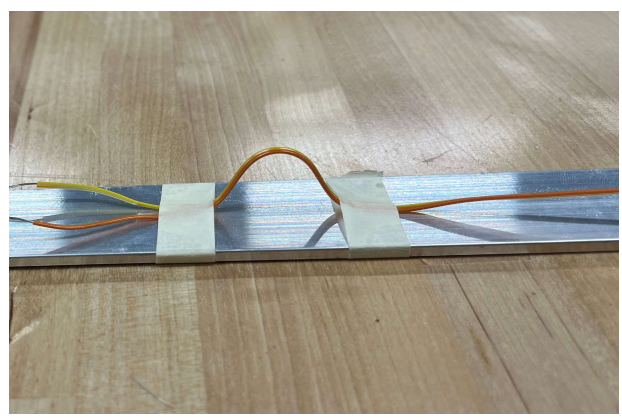

Fig. 1. Split ends of wire bent upwards and strain relief hump secured by masking tape.

The stripped ends of the wires bent upward had to be connected to the leads of the strain gauge by using a solder iron. The solder iron was heated up and put in contact with solder wire to obtain molten metal at the tip. Using tweezers in one hand to hold the strain gauge lead to the wire tip, the molten metal on the solder iron was applied to the wire-lead connection

point using the other hand. Once the solder iron was removed, both strain gauge leads were connected to the separated wire tips through the hardened solder.

To check that the strain gauge application and wire soldering were done correctly, a multimeter was used to test the resistance of the strain gauge install [3]. The free ends of the wires were clipped to the multimeter probes, and the device measured a resistance reading approximately equal to 120.0 Ohms. This was the expected equivalent resistance for the strain gauge used, therefore confirming the strain gauge and solder had been correctly applied. The final setup step was to put the aluminum bar into the mount to turn it into a cantilever beam. A screwdriver was used to loosen the mount screws so that it could be adjusted to fit the bar. The short end of the bar leading up to the strain gauge was inserted into the mount. Once the back of the bar was aligned with the back of the mount, the screwdriver was used to tighten the mount to the bar, creating the final instrumented cantilever beam whose fixed end is shown in Fig. 2 below.

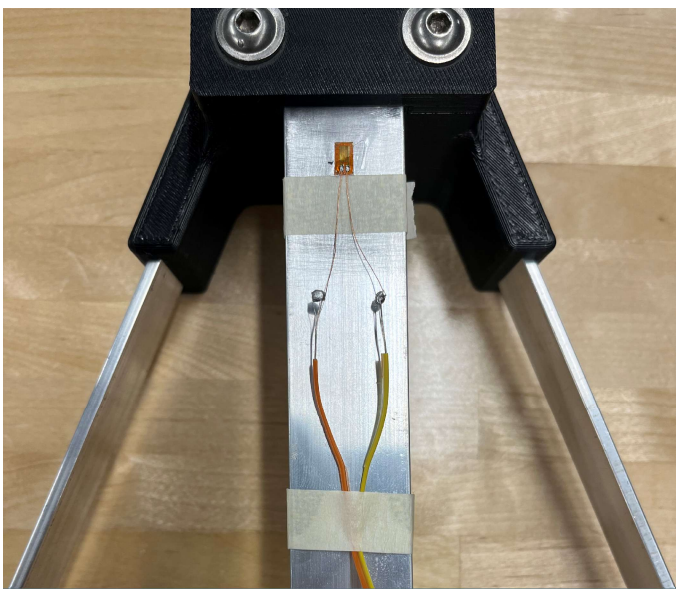

Fig. 2. The fixed end of the cantilever beam with the strain gauge leads soldered to the separated wires.

Experimental Procedure

If not already done before lab, the WSB and amplifier had to be wired to the DAQ. To connect V_s to channel 0, terminal AI0+ and AI0- on the DAQ were connected with jumper wires to terminals B+ and B- on the WSB, respectively [6]. Terminals AI1+ and AI1- on the DAQ were wired to LN+ and LN- on the WSB to connect V_g to channel 1 [6]. Terminals AI2+ and AI2- on the DAQ were connected to OUTB and OUTA on the WSB, to wire V_{amp} to channel 2 [6]. Finally, the 3.3V and GND terminals on the DAQ were connected to the VIN and GND terminals on the amplifier.

The first part of the experiment was to find the calibration constant using brass slotted weights. Four weights of similar mass not exceeding 400 grams were selected and measured in Newtons on a digital scale. The weights were incrementally added to the scale and their summed weight was recorded. This method produced a lower uncertainty than if the weights had been weighed separately and then added together after. The weights were then brought to the cantilever beam and the LabVIEW program was powered on. The free ends of the wires soldered to the strain gauge leads were connected to the strain gauge terminals on the amplifier and the DAQ was connected to the computer with a USBC cord. The program was initiated and the trim potentiometer was adjusted until the magnitude of the mean V_{amp} was less than or equal to 0.009 volts. Once this value was achieved, it was then copied and pasted into the tare control. This taring operation of V_{amp} is illustrated by equation (11) below where ΔV_{amp} is the change in amplified voltage, V_{amp} is the mean amplified voltage, and $V_{amp,tare}$ is the tare value being subtracted.

$$\Delta V_{amp} = V_{amp} - V_{amp,tare} \tag{11}$$

After taring the mean V_{amp}, the correct sample windows had to be set. The ±2.56 volts sample window was chosen for channel 0 since that was the smallest window that contained the value of V_s equal to approximately 2.50 volts. The full stack of weights was then placed on the bar and the value of mean V_{amp} was observed and the smallest sample window containing that value (±2.56 V) was chosen for channel 2 and the weight was removed. Now that sample windows were chosen, an initial point with zero loaded weight was taken. Then, the first slotted weight was loaded on the bean with the slot horizontally aligned with the tick mark that was penciled on the bar in the set up phase. The weight was slightly off centered with the slotted side further away from the midline of the bar to account for the shifted center of mass of the weight. Once the weight was placed correctly, a data point was taken. The remaining weights were added incrementally, one aligned on top of the other, with a data point being taken for each. Once the last weight was added and the final data point was taken, the program was terminated, and the data was saved to a file with a .xls filename extension.

Within the Excel file, a column was added with the mass of each weight measured by the digital scale. These digital scale measurements were then plotted on the y axis of a graph with the corresponding ΔV_{amp} values from the LabVIEW program on the x axis. A trendline was added to the graph and the option to display the equation was selected. The calibration constant was equal to the slope of the trendline shown in the equation.

With the calibration constant now determined, the weight of the can could be measured with both methods. The initial weight of the can had to be measured before any gulps were taken. The mass of the unopened can was first measured on the digital scale to serve as a baseline. The can was then brought over to the beam scale and the LabVIEW program was initiated. The same taring procedure for mean V_{amp} was performed and the calibration constant was typed into its respective control. To determine the channel 2 sample window for the soda can measurements, the full can was placed on the beam scale, and the resultant mean V_{amp} was observed to be between ±1.28 volts and ±2.56 volts before removing it. Therefore, the sample window was set to ±2.56 volts to avoid saturation and maximize resolution. The full can was then placed back on the beam aligned with the tick mark and a data point was saved. The can was removed and then placed in the same area on the beam again, and another data point was saved. This process was

repeated until ten data points for the initial weight of the can was taken. Each time a data point was saved, the weight of the can from both the beam scale method and the calibration method were recorded.

After the initial weight measurements were taken, the LabVIEW program was reset, the same taring process was performed, and the calibration constant was re-entered if not previously saved. The full can was opened and placed on the beam aligned with the tick mark, as shown in Fig. 3 below, and a data point was saved.

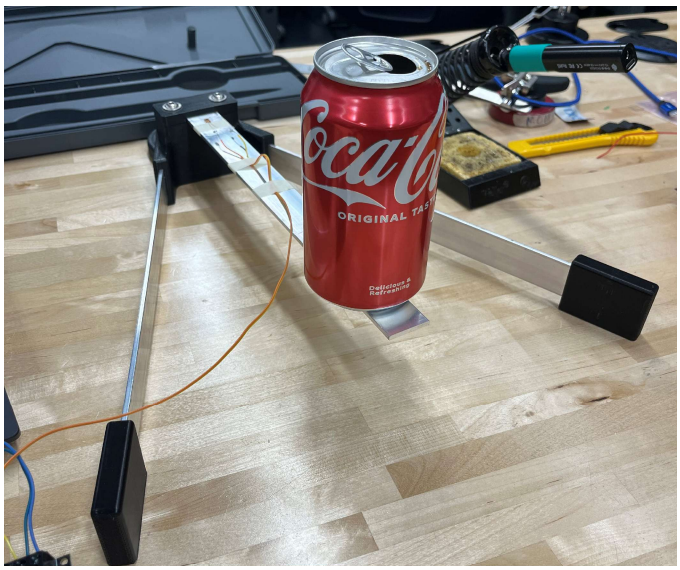

Fig. 3. Opened soda can correctly placed on instrumented cantilever beam during gulp experiment.

Next, the can was removed, a gulp was taken from it, and then it was placed back on the beam in the same position. Another data point was saved, and this process of taking gulps from the can and saving data was repeated until the can was empty. A final data point was taken for the empty can on the beam and then the LabVIEW program was terminated and the data was saved to a file with a .xls extension. The weight of the empty can was then measured on the digital scale, and all trash, including the empty can, was disposed of.

III. Results

The dimensions of the cantilever beam and the soda can measured using digital instruments are shown in Table I below. These measurements include the width of the beam, b, the height of the beam, h, the full length of the beam, L_0, the length from the center of the strain gauge to the free end of the beam minus half the can diameter, L, and the diameter of the can, d.

Table I
Dimensions of Cantilever Beam and Soda Can

Dimension	Measurement (m)
b	0.02498 ± 0.000005
h	0.002961 ± 0.000001
L	0.23678 ± 0.0005
L_0	0.26826 ± 0.0005
d	0.03296 ± 0.000005

To find the calibration constant, four brass slotted weights were incrementally measured on the digital scale in Newtons and the total weight after each increment was recorded. The weights were then loaded on the cantilever beam in the same fashion and the mean amplified voltage was recorded for each increment. The weight measurements from the digital scale, P, and their associated mean amplified voltage measured from the beam, V_{amp}, are shown in Table II below. The calibration constant, C, found by taking the slope of the trendline of the data below was equal to 8.5138 ± 0.7487 N/V.

Table II
Calibration Weights and Voltages

P [N]	V_{amp} [V]
0.0000 ± 0.0005	0.0031 ± 0.0077
0.4847 ± 0.0005	0.0581 ± 0.0081
0.9737 ± 0.0005	0.1132 ± 0.0099
1.4657 ± 0.0005	0.1732 ± 0.0116
1.9570 ± 0.0005	0.2329 ± 0.0231

For the unopened can initial weight measurements, ten trials were taken on the instrumented cantilever beam and the weight was recorded using the beam scale method, W, and the calibration method, W_c. The associated voltage, ΔV_{amp}, was also recorded for each of the ten measurements. These three parameters for the ten initial weight measurements are shown in Table III below.

Table III
Soda Can Initial Weight Measurement Data

ΔV_{amp} [V]	W [N]	W_c [N]
0.4404 ± 0.0199	3.993 ± 0.1922	3.749 ± 0.3679
0.4469 ± 0.0226	4.052 ± 0.2145	3.805 ± 0.3832
0.4522 ± 0.0161	4.100 ± 0.1634	3.850 ± 0.3622
0.4487 ± 0.0277	4.069 ± 0.2569	3.820 ± 0.4079
0.4495 ± 0.0353	4.076 ± 0.3204	3.827 ± 0.4485
0.4313 ± 0.0166	3.911 ± 0.1652	3.672 ± 0.3496
0.4422 ± 0.0164	4.010 ± 0.1644	3.765 ± 0.3562
0.4367 ± 0.0241	3.960 ± 0.2261	3.718 ± 0.3834
0.4320 ± 0.0204	3.917 ± 0.1951	3.678 ± 0.3641
0.4439 ± 0.0231	4.025 ± 0.2183	3.779 ± 0.3834

After the initial weight measurements, incremental gulps were taken from the soda can with a data point being saved after each one. The weight of the can after each gulp from both methods and the associated amplified voltages are shown in Table IV below.

Table IV
Weight of Can After Each Gulp

ΔV_{amp} [V]	W [N]	W_c [N]
0.4277 ± 0.0144	3.878 ± 0.1479	3.641 ± 0.3424
0.3992 ± 0.0129	3.620 ± 0.1341	3.399 ± 0.3181
0.3652 ± 0.0126	3.312 ± 0.1285	3.110 ± 0.2933
0.3316 ± 0.0131	3.007 ± 0.1297	2.823 ± 0.2719

Table IV Continued
Weight of Can After Each Gulp

ΔV_{amp} [V]	W [N]	W_c [N]
0.2792 ± 0.0144	2.532 ± 0.1359	2.377 ± 0.2420
0.2521 ± 0.0127	2.286 ± 0.1202	2.146 ± 0.2172
0.2231 ± 0.0166	2.023 ± 0.1512	1.900 ± 0.2185
0.1950 ± 0.0260	1.768 ± 0.2312	1.660 ± 0.2649
0.1756 ± 0.0144	1.592 ± 0.1301	1.495 ± 0.1793
0.1492 ± 0.0129	1.353 ± 0.1162	1.271 ± 0.1563
0.1265 ± 0.0131	1.147 ± 0.1171	1.077 ± 0.1460
0.0961 ± 0.0128	0.871 ± 0.1136	0.818 ± 0.1303
0.0643 ± 0.0127	0.583 ± 0.1126	0.548 ± 0.1187
0.0302 ± 0.0128	0.274 ± 0.1131	0.258 ± 0.1116

The size of each gulp was determined by taking the difference of the weight of the can between each increment. The results for gulp size from both methods, W_g and W_{cg}, along with the number of gulps, N_g, is shown in Table V below.

Table V
Gulp Size Data

N_g	W_g [N]	W_{cg} [N]
1	0.2578	0.2421
2	0.3083	0.2894
3	0.3054	0.2867
4	0.4751	0.4460
5	0.2458	0.2308
6	0.2623	0.2463
7	0.2553	0.2397
8	0.1758	0.1650
9	0.2392	0.2246
10	0.2060	0.1934
11	0.2757	0.2589
12	0.2881	0.2705
13	0.3091	0.2902

The additional given parameters needed to calculate weight using the beam scale method including the modulus of elasticity, E, the gauge factor, G_f, and the gain for voltage amplification, *gain*, are provided in Table VI below. Also included is the yield stress of aluminum 6061-T6, σ_y, found from an online source [1].

Table VI
Given Parameters for Beam Scale Method and Yield Stress

Parameter	Value	Units
E	69.0 ± 0.09	GPa
G_f	2.085 ± 0.0104	none
gain	900 ± 4.50	none
σ_y	276 ± 0.60	MPa

The weight of the full soda can and the weight of the empty soda can were measured on a digital scale to serve as a baseline. The results for these digital scale measurements are shown in Table VII.

Table VII
Weight of Full and Empty Can on Digital Scale

Full or Empty Weight	Value [N]
W_{full}	3.7905 ± 0.0005
W_{empty}	0.1514 ± 0.0005

Finally, the tare values for the amplified voltage are shown in Table VIII below. These include the tare value from finding the calibration constant, $V_{amp,tare,1}$, the tare value from taking the initial weight of the can measurements, $V_{amp,tare,2}$, and the tare value from the gulp data, $V_{amp,tare,3}$.

Table VIII
Amplified Voltage Tare Values

$V_{amp,tare,\#}$	Value [V]
$V_{amp,tare,1}$	0.0028 ± 0.0077
$V_{amp,tare,2}$	0.0113 ± 0.0138
$V_{amp,tare,3}$	0.0078 ± 0.0099

IV. Discussion

This experiment utilized two separate methods to measure the weight of a soda can on an instrumented cantilever beam. The beam scale method relied on mechanics of materials principles to calculate the weight of the can on the beam using (9). The calibration method utilized a constant multiple relationship with the amplified voltage shown in (10) to determine the weight of the can on the beam. Although both methods generated similar results, differences in their uncertainties provided insight towards which one was more accurate and precise.

Interpretation of Results and Analysis of Data

The first part of the experimental procedure was to use brass slotted weights to determine the calibration constant. Table II shows the weights measured on the digital scale and the amplified voltage produced when these weights were placed on the cantilever beam. The relationship between these two parameters is better shown by Fig. 4 showcasing the trendline and equation used to find the calibration constant.

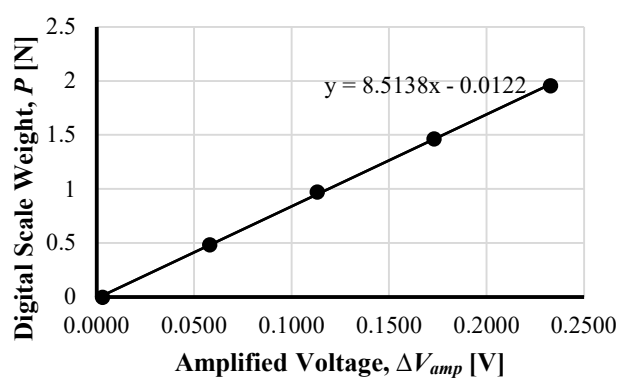

Fig. 4. Digital scale weight plotted against amplified voltage to determine calibration constant.

It can be observed that all points are in contact with the trendline, suggesting a naturally linear relationship between the parameters. These same five data points plotted above were used in a five-point Monte Carlo simulation to determine the uncertainty of the calibration constant.

The ten initial weight measurements for the full can are shown in Table III for both methods. The mean of the initial weight measurements from the beam scale method, $mean_W$, and the calibration method, $mean_{Wc}$, were equal to 4.01 ± 0.15 N and 3.77 ± 0.14 N, respectively. The uncertainties were determined by taking the standard deviation of the data from each method and then multiplying by a t number to get the 95% confidence interval. Since the full can was measured ten times, the t number corresponding to a sample size of 10 was chosen which had a value equal to 2.262 [5]. A comparison to the digital scale weight of the full soda can in Table VII reveals that the calibration constant was more accurate at measuring the weight of the full can. The mean value of the full can weight from the calibration method was nearly equal to the digital scale measurement, while the mean weight from the beam scale method was more than 0.20 Newtons higher than the digital scale measurement.

The decrease in the weight of the can as gulps were taken is shown in Table IV for both methods. The graph below in Fig. 5 plots this data to show that the relationship between the number of gulps and the weight of the can was roughly linear.

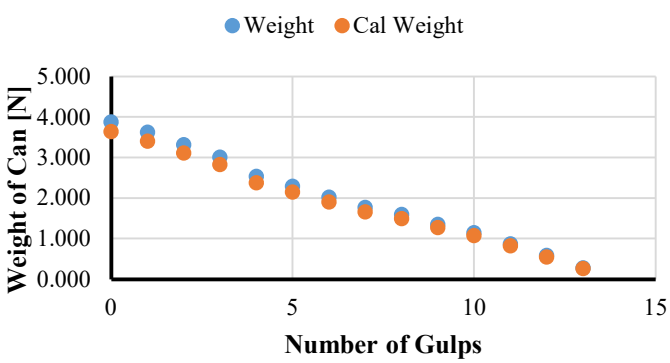

Fig. 5. Weight of soda can versus number of gulps for beam scale method and calibration method.

The number of gulps begins at zero because the first data point was taken for when the can was still full. It can be observed that the calibration method weight was consistently lower than the weight of the can approximated by the beam scale method. However, as more gulps were taken, the weight recorded by both methods began to converge and get closer together in value. By the final gulp, the values for the weight of the can were nearly equal as shown by the full overlap of the thirteenth data point above.

Although the slope of the graph in Fig. 5 looks roughly linear, plotting the weight of each individual gulp reveals the decreases in can weight are not as constant as they appear as shown in Fig. 6.

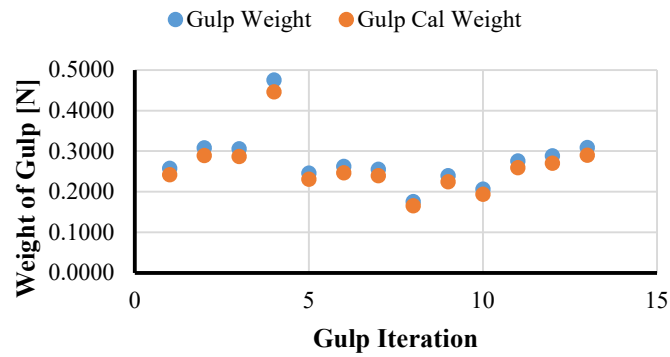

Fig. 6. Weight of each gulp from beam scale method and calibration method.

While most of the gulps fell within the range of 0.20 to 0.30 Newtons, there was one minor outlier weighing below 0.20 Newtons and one major outlier weighing above 0.40 Newtons. The exact values of these outliers can be observed in Table V. The reason behind this inconsistency in gulp size was due to the lack of precision associated with a human taking gulps out of a soda can. The human mouth cannot function like a pipette by precisely measuring the amount of liquid inside it, and therefore the gulp sizes were inconsistent. Based off the gulp size data, the mean gulp size and uncertainty could be calculated.

Mean Gulp Size and Uncertainty

The mean gulp size was found by adding up all the individual gulp weights and then dividing their sum by the total number of gulps which in this case was equal to 13. The mean gulp size for the beam scale method, $mean_{Wg}$, was equal to 0.28 ± 0.16 N and the mean gulp size for the calibration method, $mean_{Wgc}$, was equal to 0.26 ± 0.15 N. The uncertainty for these mean values were determined by taking the standard deviation of the gulp sizes and then multiplying that by the t number. The t number was selected based on the number of measurements, which in this case was equal to the number of gulps taken. A standard t number was not provided for a sample size of 13 [5]. However, a t number equal to 2.262 was provided for 10 measurements and a t number equal to 2.045 was provided for 30 measurements [5]. Therefore, a linear interpolation was performed between these two points to approximate the t number for 13 measurements. The interpolation generated a t number equal to 2.229, and this value was multiplied by the standard deviation of the gulp sizes found using an Excel function to determine the uncertainty of the mean gulp size for both methods.

The uncertainty for the mean gulp size of the calibration method was slightly less than the uncertainty generated from the beam scale method. This makes sense as the gulp sizes generated from the calibration method deviate more from the beam scale method for the larger gulp sizes. For example, in Fig. 6, the calibration method weight for the largest gulp (fourth point) was more proportionally decreased from the beam scale method weight than it was for the smallest gulp size (eighth point). Essentially, the calibration method weight was lower than the beam scale weight at both points, but there was

significantly more overlap between each method at the lower gulp weight showing a smaller difference. Therefore, the range of gulp size for the calibration method was less than that of the beam scale method leading to a higher overall precision and lower uncertainty.

Comparison of Methods to Determine the Best

As previously discussed, the calibration method more accurately approximated the weight of the full can when compared to the digital scale method and it produced a lower uncertainty for the mean gulp size. However, these are not the only two things to be considered for choosing the best method. For the empty can, the weight from the calibration method was also closer to the digital scale measurement than the beam scale method, but by a much smaller margin. Therefore, the calibration method was more accurate than the beam scale method when compared to the digital scale measurements. However, the uncertainties generated for the weight of the can as gulps were taken from it were consistently larger for the calibration method than the beam scale method as shown in Table IV. This suggests that although more accurate in this experiment, the calibration method is overall less precise than the beam scale method due to its larger uncertainties. If the experiment was repeated multiple times with new calibration constants being calculated each time, then the weights generated from this method will likely vary more between experiments than the weights from the beam scale method.

While the calibration method is more accurate and the beam scale method is more precise, both these methods are limited by the yield stress of the aluminum beam. To determine the maximum weight the beam can measure, W_{max}, equation (5) was first rearranged to solve for W. Then, the yield stress, σ_y, was plugged in for σ_x, and the full length of the beam, L_0, was plugged in for L, leading to (12) shown below.

$$W_{max} = \sigma_y \frac{bh^2}{6L_0} \tag{12}$$

Plugging in the respective values from Table I and Table VI, yields a maximum weight of 37.55 ± 0.080 N. In this experiment, the heaviest weight placed on the beam scale was the full soda can which weighed 3.7905 ± 0.0005 N according to the digital scale. Therefore, the beam did not come close to yielding as the weight of the can was far less than the maximum weight calculated from the yield stress. However, if this maximum was exceeded in a different experiment, then the beam would fail on the top surface near the clamp and the accuracy of the resulting measurements would be diminished.

Overall, it was difficult to choose the best method based on this experiment. An analysis of the results reveals that the beam scale method will generate more precise results for the weight of the can as gulps are taken while the calibration method will produce more accurate results. Both methods are limited by the maximum weight the beam can handle before yielding, and therefore, if this weight is to be exceeded then a new method will need to be developed.

Limitations of this Experiment

There were multiple limitations that affected the results of this experiment. First off, there was human error in where the can was placed on the beam. Although a tick mark was made on the beam where the can and weights were supposed to be placed, it was difficult to align the objects on the tick mark perfectly every time. This was especially true for the can as it would completely cover the tick mark making it difficult to center correctly. These variations in can placement affected the weight measurements produced by the beam scale method due to the dependence of moment, M, on length, L.

Next, thermal strain could have occurred from resistors heating up if the experiment had a long duration. This additional strain could have impacted results by increasing the weights measured by both methods. Another limitation of this experiment was the difficulty in measuring small weights on the cantilever beam such as the empty can. The sample windows were set based on the weight of the full soda can. Therefore, when the empty can was measured, the resolution was not fine enough for the DAQ to measure a highly accurate voltage reading.

As previously mentioned, the methods were also limited by the maximum weight the cantilever beam could handle before yielding. However, this did not impact the objectives of this specific experiment as no weights exceeding this value were used. The weight of the beam affecting calibration and measurements was not a limitation of this experiment. The reason for this is that taring the amplified voltage before taking data negated the effects of the beam weight.

Suggestions for Improvement

To improve the consistency of the placement of the soda can, the circumference of the bottom of the can could be traced on the beam at the proper location. The can could then be placed directly aligned with the arc traced from its circumference, rather than trying to align with a small tick mark that is blocked from the size of the can.

The effects of thermal strain could be better negated by performing the experiment in a cooler setting and by taking longer breaks between the parts of the experiment that rely on the strain gauge. To improve the ability of the methods to measure small weights, a half WSB could be used instead of a quarter bridge and the length of the beam could be doubled. Both these changes would double the WSB voltage, V_g, which in turn would double the amplified voltage, V_{amp}. This would increase the voltage readings produced by smaller weights allowing for a more accurate measurement at the same resolution.

V. Conclusion

The mechanics of material derived beam scale method was compared to the calibration method by weighing a soda can on an instrumented cantilever beam. The weight of the full can was first measured ten times on the cantilever beam. The beam scale method recorded a mean weight of 4.01 ± 0.15 N for the full can while the calibration method recorded a mean weight of

3.77 ± 0.14 N. A comparison to the digital scale weight of the full soda can revealed that the calibration method was more accurate. Thirteen gulps were then taken to empty the can with the weight being recorded by both methods at every increment. The mean gulp size from the beam scale method was equal to 0.28 ± 0.16 N while the mean from the calibration method was equal to 0.26 ± 0.15 N. Although the calibration method was deemed more accurate, the average gulp size and associated uncertainty between the methods were similar. An uncertainty analysis of the weight of the can after each gulp increment revealed that the calibration method was less precise than the beam scale method as it had a much greater uncertainty associated with each individual weight measurement.

Limitations to this experiment included inconsistent can placement, thermal strain, and inaccuracies in measuring small voltage values. Additionally, the amount of weight used in a similar experiment would be limited by the yield stress of 6061 T6 aluminum and the associated maximum weight that the cantilever beam could handle before yielding. Improvements such as tracing the bottom can circumference on the beam, performing the experiment in a cooler environment with more breaks, using a half bridge or doubling the length of the beam to increase WSB voltage, and selecting a stronger material could be made to overcome the previously stated limitations.

APPENDIX

All parameters used in this experiment had an associated uncertainty. Four different methods of determining uncertainty were utilized depending on how the parameter was measured or calculated. First, some uncertainties were provided by manufacturers or known by rule of thumb (Table IX). Statistical uncertainties with a 95% confidence interval were found for the tared voltage values based off 1000 samples and for the mean initial weight of can and mean gulp size calculations based off their respective sample sizes (Table X). The uncertainties for parameters dependent on other variables were found using RSS propagation equations that considered all contributing uncertainties (Table XI). Finally, a five-point Monte Carlo simulation was used to calculate the uncertainty of the calibration constant. The uncertainties of the five amplified voltage and digital scale weight values used in the simulation as well as the uncertainty of the calibration constant are detailed in Table XII.

Table IX
Given Uncertainties

Parameter (units)	μ	Uncertainty	Source
P (N)	μ_P	± 0.0005	Manufacturer [2]
b (mm)	μ_b	± 0.000005	Manufacturer [4]
h (mm)	μ_h	± 0.000001	Manufacturer [4]
L (mm)	μ_L	± 0.0005	Rule of Thumb [4]

Table IX Continued
Given Uncertainties

Parameter (units)	μ	Uncertainty	Source
L_0 (mm)	μ_{L0}	± 0.0005	Rule of Thumb [4]
d (mm)	μ_d	± 0.000005	Manufacturer [4]
W_{full} (N)	μ_{Wfull}	± 0.0005	Manufacturer [2]
W_{empty} (N)	μ_{Wempty}	± 0.0005	Manufacturer [2]
G_f (none)	μ_{Gf}	± 0.0104	Manufacturer [5]
$gain$ (none)	μ_{gain}	± 4.50	Manufacturer [5]

Table X
Statistical Uncertainties

Parameter (units)	μ	Uncertainty	# Samples
$V_{amp,tare,1}$ (V)	$\mu_{Vamp,tare,1}$	± 0.0077	1000
$V_{amp,tare,2}$ (V)	$\mu_{Vamp,tare,2}$	± 0.0138	1000
$V_{amp,tare,3}$ (V)	$\mu_{Vamp,tare,3}$	± 0.0099	1000
$mean_W$ (N)	μ_{meanW}	± 0.15	10
$mean_{Wc}$ (N)	μ_{meanWc}	± 0.14	10
$mean_{Wg}$ (N)	μ_{meanWg}	± 0.16	13
$mean_{Wgc}$ (N)	$\mu_{meanWgc}$	± 0.15	13

Table XI
RSS Propagated Uncertainties

Parameter	μ	Uncertainty	Units
ΔV_{amp}	$\mu_{\Delta Vamp}$	± 0.0253	V
W	μ_W	± 0.1682	N
W_c	μ_{Wc}	± 0.2840	N
σ_y	$\mu_{\sigma y}$	± 0.60	MPa

The uncertainty for every ΔV_{amp}, W, and W_c measurement was calculated with the respective RSS propagation equations separately shown in Table III and Table IV. The uncertainties for these parameters displayed in the table above is an average of all the separate uncertainties as the values varied.

Table XII
Monte Carlo Simulation Uncertainties

Parameter	μ	Value and Uncertainty	Units
V_{amp1}	μ_{Vamp1}	0.0031 ± 0.0077	V
V_{amp2}	μ_{Vamp2}	0.0581 ± 0.0081	V
V_{amp3}	μ_{Vamp3}	0.1132 ± 0.0099	V
V_{amp4}	μ_{Vamp4}	0.1732 ± 0.0116	V
V_{amp5}	μ_{Vamp5}	0.2329 ± 0.0231	V
P_1	μ_{P1}	0.0000 ± 0.0005	N
P_2	μ_{P2}	0.4847 ± 0.0005	N
P_3	μ_{P3}	0.9737 ± 0.0005	N
P_4	μ_{P4}	1.4657 ± 0.0005	N
P_5	μ_{P5}	1.9570 ± 0.0005	N
C	μ_C	8.5138 ± 0.7487	N/V

RSS Propagation Uncertainty Equations:

RSS Propagation of (11) for ΔV_{amp}

$$\mu_{\Delta V_{amp}} = [\frac{\partial \Delta V_{amp}^2}{\partial V_{amp}} \mu_{Vamp}^2 + \frac{\partial \Delta V_{amp}^2}{\partial V_{amp,tare}} \mu_{Vamp,tare}^2]^{1/2} \quad (13)$$

RSS Propagation of (9) for W

$$\mu_W = [\frac{\partial W^2}{\partial \Delta V_{amp}} \mu_{\Delta V_{amp}}^2 + \frac{\partial W^2}{\partial gain} \mu_{gain}^2 + \frac{\partial W^2}{\partial V_s} \mu_{V_s}^2 + \frac{\partial W^2}{\partial G_f} \mu_{G_f}^2 + \frac{\partial W^2}{\partial \mathrm{E}} \mu_E^2 + \frac{\partial W^2}{\partial b} \mu_b^2 + \frac{\partial W^2}{\partial h} \mu_h^2 + \frac{\partial W^2}{\partial L} \mu_L^2]^{1/2} \quad (14)$$

RSS Propagation of (10) for W_c

$$\mu_{Wc} = [\frac{\partial Wc^2}{\partial \mathrm{C}} \mu_C^2 + \frac{\partial Wc^2}{\partial \Delta V_{amp}} \mu_{\Delta V_{amp}}^2]^{1/2} \quad (15)$$

RSS Propagation of (12) for W_{max}

$$\mu_{W_{max}} = [\frac{\partial {W_{max}}^2}{\partial \sigma_y} \mu_{\sigma_y}^2 + \frac{\partial {W_{max}}^2}{\partial b} \mu_b^2 + \frac{\partial {W_{max}}^2}{\partial h} \mu_h^2 + \frac{\partial {W_{max}}^2}{\partial L_0} \mu_{L_0}^2]^{1/2} \quad (16)$$

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